

Exploring Descript Audio Codec's latent space

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Colab link:

<https://colab.research.google.com/github/inspektral/probingDAC/blob/main/probingDAC.ipynb>

Github repository: <https://github.com/inspektral/probingDAC>

Video: <https://youtu.be/Z9sBE4-B6vw>

Abstract

Descript Audio Codec (DAC) is a high-fidelity universal audio encoder based on neural compression and discrete tokenization. Its ease of use, quality, and universality make it a very interesting tool for generative and more creative tasks, since its latent space is very dense with information learned from recordings of all kinds, and even random walks within it produce interesting results. The problem with this is that being intentional and finding specific sounds, variations, and morphing trajectories, given its compression-oriented nature, is neither intuitive nor easy. This project is a small exploration of DAC's latent space in order to find some mechanics trajectories and build a framework of intuition to play with it more intentionally.

Introduction

In recent times, latent spaces are being explored and used for sound art, the most well known model for this is RAVE and its recent diffusion based version AFTER, those are all VAE based and the idea is that they are trained on a small and specific dataset (e.g. guitar, spoken word) and have a very structured latent space to perform morphs, modulations style transfer in real time and with great audio quality. The only problem is the dataset, and under this point of view neural encoders made for compression purposes are very interesting because they are trained on a huge variety of sounds, in order to be able to compress any kind of sound. In this field DAC is one of the most advanced, it is inspired by GAN architecture of SoundStream, it is trained on spoken voice, music and ambient sounds and it is able to compress any sound by a factor of 90 with unperceivable loss. Since the DAC is not made for be manipulated in the latent space it the space is much harder to understand and this work is a preliminary collection of experiments that aims at building a framework to test this and other latent spaces and build an intuition to play with them.

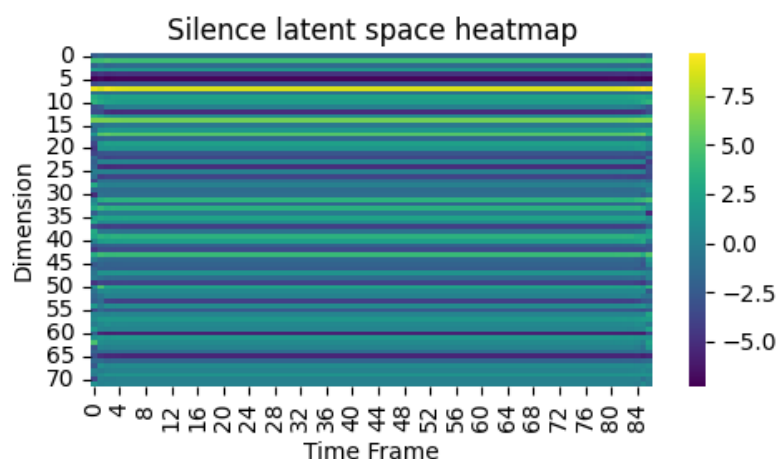
Methods and experiments

To explore the latent space of the Descript Audio Codec (DAC), several experiments were conducted using different methodological approaches. Initially, the project involved the analysis of simple, one-second-long sounds, specifically silence, a 1000Hz sine wave, and white noise, by plotting their heatmaps in the latent space to observe their representations. Following this, quantitative analyses were performed by measuring the average distance in the latent space: first, between silence and sine waves at various frequencies, and subsequently, between a 1000Hz sine wave and silence at different amplitude levels. Furthermore, the research included creative exploration through direct manipulation of the latent space dimensions. This involved "gating noise" by setting most dimensions to the values corresponding to silence while assigning noise values to only a few dimensions. Another creative experiment focused on distorting a 200Hz sine wave by selectively amplifying its first few latent dimensions. Finally, a series of experiments investigated the possibility of synthesizing pitched sounds by injecting sine waves into individual latent dimensions in various ways, including using single sine waves, sine waves with their harmonics, phase-shifted copies of sine waves, and random sine waves. The resulting audio was then analyzed to understand the relationship between these manipulations and the produced sounds.

Simple sounds

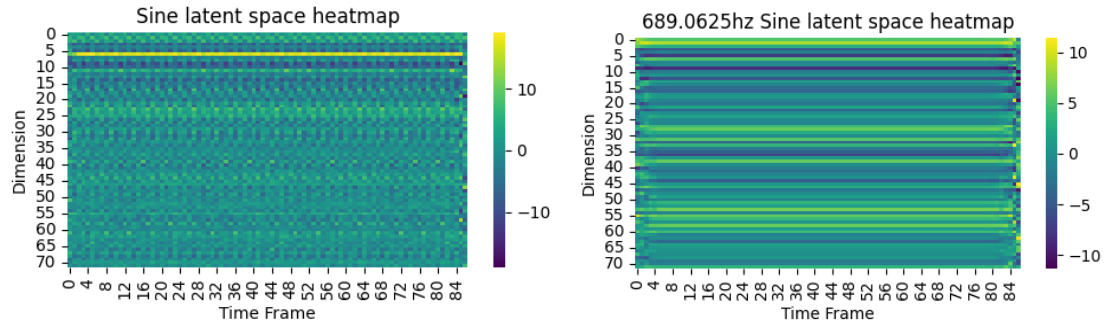
The first experiment that was tried was plotting the heatmap of 3 1 second long simple sounds: silence, 1000hz sine wave and white noise. The expected results were that silence would be constant, sine would either be constant or exhibit some periodic behaviour and noise would be essentially random. Actual results matched the expected as follows.

Silence



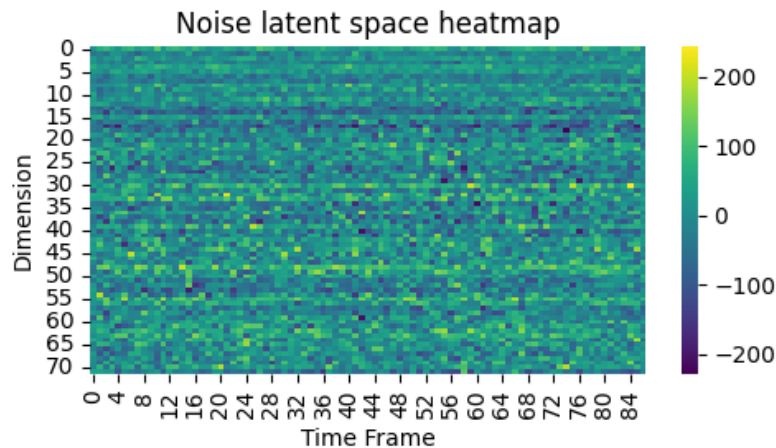
As expected silence is constant, but it is nowhere near the zero of the space, there's no reason for it to be there so this had to be expected but it is nonetheless worth mentioning and it is taken into account in the other experiments of this report.

Sine wave



The 1000hz sine wave exhibits a periodic behaviour, this is to be attributed to how phase and windowing interact, on the other hand a 689.0625hz (sampling_rate/encoder_dimension) sine and its harmonics and subharmonics are stable. This result is almost obvious, but sadly it makes synthesis by manipulating this latent representation quite difficult. Further investigation of the FFT spectrum of this representation could lead to interesting results, but has not been done in this preliminary series of experiments.

White noise

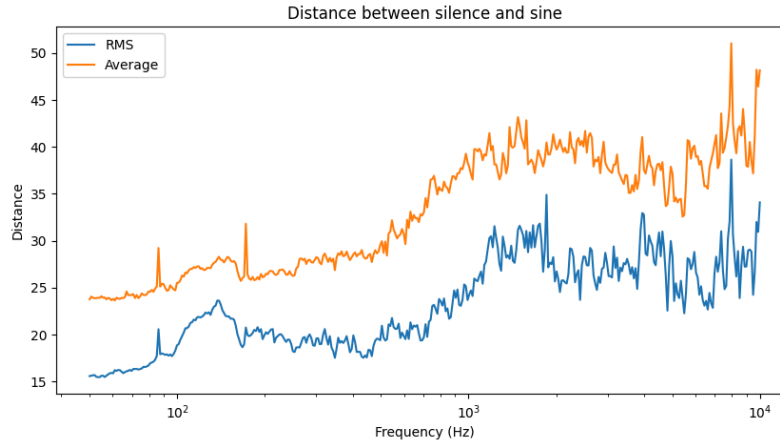


As expected noise is quite random, but when listened to it has some wordlike features, transients and formants surely learned from the dataset; it could be interesting to iteratively encode and decode to see if it converges to some original training samples or a combination of those. Differently from expectation the size of those vectors is an order of magnitude higher than the sines. This is interesting, since white noise is the most discontinuous and a sine wave is very continuous, scaling vectors in the latent space could lead to more discontinuous, noisy and harmonically rich sounds. Something like clipping in the latent space.

Unfortunately this has been tried and scaling all vectors by the same factor doesn't change the sound (when normalized), which doesn't really explain the cause of this magnitude for noise. Direction seems to be the important part of the vector, and scaling can be applied to a subset of dimensions to act on that in interesting ways. In the same way scaling down doesn't work in a creative denoising fashion.

Frequencies

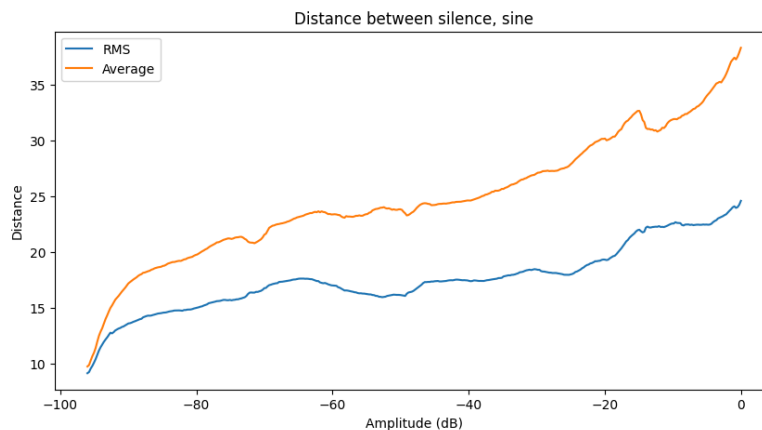
The next experiment, given that sines exhibit a periodic behaviour in the latent space, is the evaluation of the average distance between silence and sines at different frequencies, without focusing on specific dimensions.



A slight growth is found, but it is very irregular, its significance is probably negligible. The irregularity is quite interesting, especially the bump around 1-2kHz, which might be due in some way to the big amount of voice samples this model is trained on and has thus the ability to distinguish and represent.

Amplitudes

The same experiment has been done also with amplitudes, by measuring the average distance between a 1000hz sine and silence.



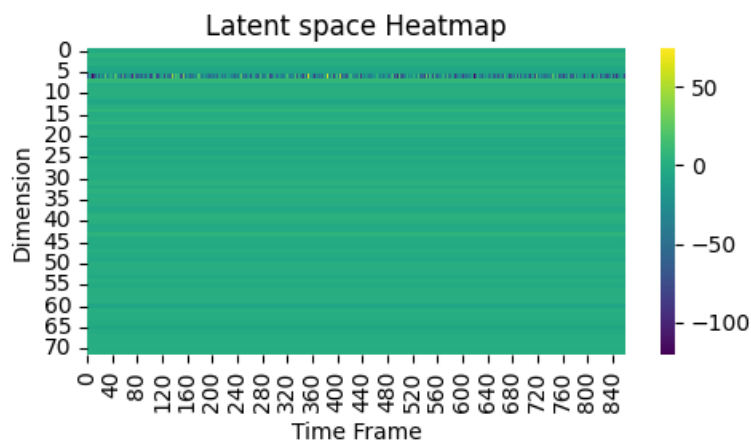
A growth is found, much more regular, almost directly proportional to the amplitude (in db).

Creative modifications and synthesis

Various creative experiments have been tried with a variety of results, most of them quite unpredictable but quite entertaining that suggest that DA's latent space can be an interesting palette to explore for various kinds of sound art.

Gating noise

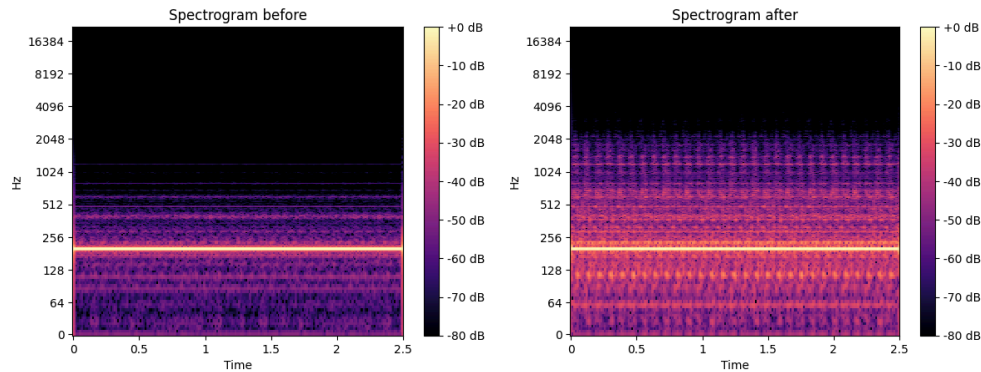
Setting every dimension of the latent space equal to the silence one and only one or a few ones to the values of noise. This leads to various kinds of crackles and textures, very organic, sometimes more metallic or woody some other times more vocal. It is also a clear example that the reduced dimensions are ordered from the most significant, tweaking the first will lead to big variations, tweaking the last probably to unperceivable ones. The fact that the model is already trained suggests that it would be very interesting to use it as a texture generator just by tweaking the distance from silence of the randomness in each channel.



Distorting sines

The second creative experiment that has been tried is the distortion of a 200hz (chosen in the lower side to give more room for harmonics and other noises), thanks to the previous experiments we know that amplifying all dimensions won't have interesting effects, but amplifying the first ones should have a very noticeable effect. The first 5 dimensions are thus scaled by a factor of 10.

The results are quite interesting, the base frequency is perfectly maintained, and the distortion is centered on that, decaying when getting away from it. It is very natural and non digital, some rhythmic patterns are introduced and it sounds very much like some sort of machinery.

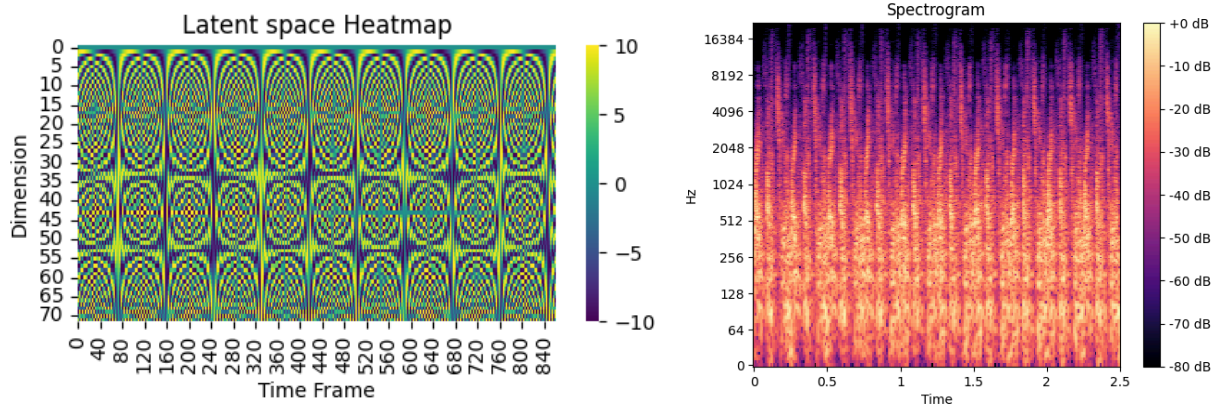


Sines in the latent space

Since periodic sounds exhibit periodic behaviour in the latent space when they're out of sync with the window the next series of experiments is about injecting sines in each dimension in various ways and to check whether it is possible to obtain easily pitched sound and perform pitch manipulations. Under the point of view the results are unsuccessful, but other interesting phenomena have been detected.

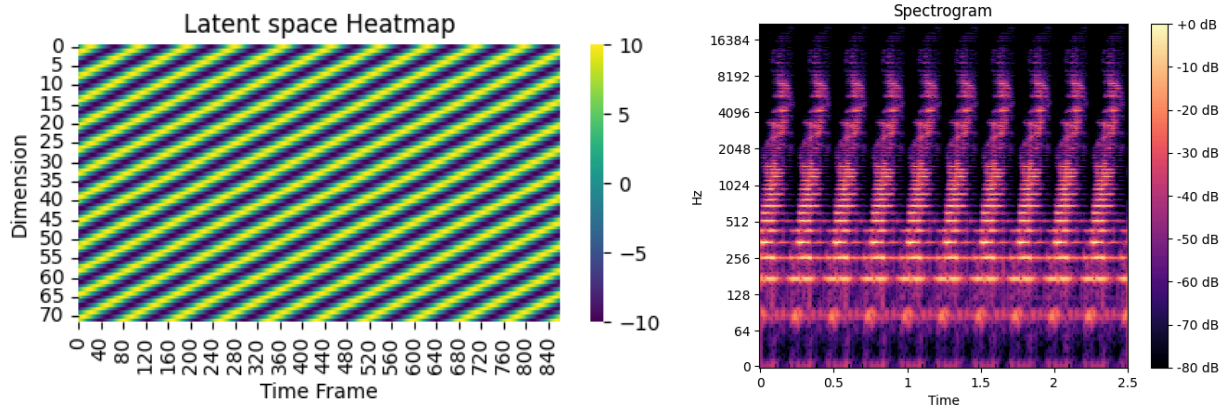
Harmonics in the latent space

For the first experiment a sine wave and its harmonics have been assigned to the dimensions in the latent space, the produced pattern is clear from the heatmap, but the result is very unexpected, it is in fact not a pitched sound, it is clearly a loop and it sounds very much like a robotic voice. The sound is quite interesting from itself but it doesn't differ much from a smooth noise.



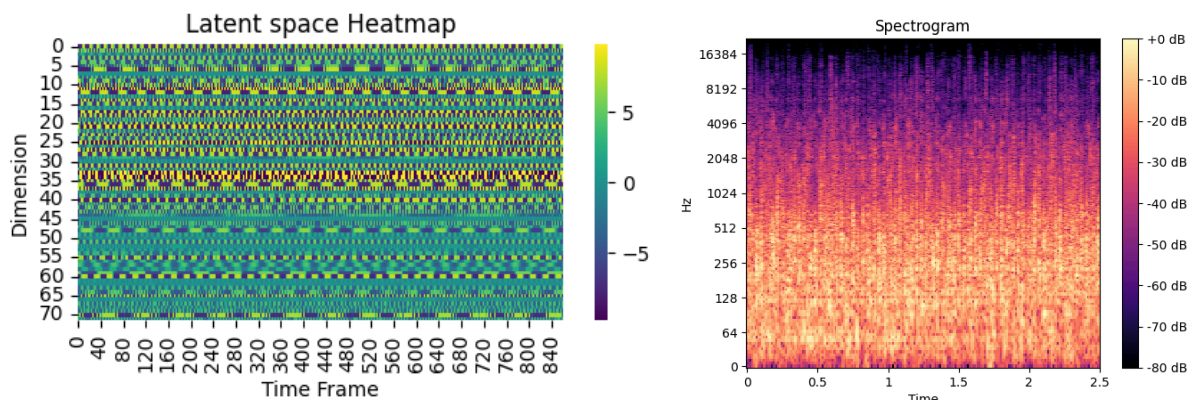
Phase shifting

For the second experiment a sine wave and phase shifted copies of itself have been assigned to the dimensions of the latent space, the result is a pitched sound on which aggressive low pass filtering is performed. It sounds rather pleasing and it could be an effect that can be applied to other sounds, maybe summing or multiplying their latent spaces by these kinds of patterns, it could be an interesting modulation, especially if done with caution in the higher dimensions.



Random sines

The third and last experiment with sines is with random ones, amplitude, frequency and phase is randomized. The result is unpitched noise, but it's quite different from white noise, it contains rhythmic patterns and voice-like sounds. From this it is quite clear that dimensions are really interdependent and must be tweaked taking this into consideration, apart from this the result is not very insightful.



Conclusion

The main outcome of those experiments is that understanding DAC's latent space is not an easy task, manipulating it with intention is even harder but it is nevertheless an interesting playground for different creative approaches, especially as an effect it behaves somewhat predictably, texture of synthesis is also viable, and knowing where silence is can work as a guide for creating complex envelopes and gestures. Creating pitched sounds with precision is very hard though, but there are many tools available (e.g. DDSP) that are perfect for that, thus it is not even that interesting, especially in the scope of this creative research. One very interesting feature is that the dataset comprehends a lot of ambient noises and a lot of ambient sounds, and wandering around with some tools to orient oneself, knowing which tweaks have more or less impact is undoubtedly useful to obtain interesting very organic morphs.

References

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